Quiz 3 - DSC 10, Spring 2025

Full Name:

PID:

Quiz Time: ◯ 3PM ◯ 3:30PM ◯ 4PM ◯ 4:30PM

Instructions:

- This quiz consists of 2 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

Question 1

The DataFrame `concerts` contains data on a sample of concerts held in 2024. For each concert, we have the name of the performing `"artist"`, the `"date"` and `"location"` of the concert, and the total `"attendance"` as an integer. The first few rows of `concerts` are shown below:

	artist	date	location	attendance
0	Drake	2024-02-16	Pittsburgh	30000
1	Conan Gray	2024-10-20	Phoenix	15000
2	Sabrina Carpenter	2024-11-10	San Diego	14000

a) You are interested in estimating the average `"attendance"` at concerts using the data in `concerts`. Fill in the blanks to define a function `estimate_attendance` that takes as input a number of estimates to produce, and returns an array with that number of bootstrapped estimates of the mean concert attendance.

```
def estimate_attendance(how_many):
    estimates = np.array([])
    for i in __(a)__:
        resample = ___(b)__
        estimates = np.append(estimates, __(c)__.mean())
    return estimates
```

(a):

(b):

(c):

b) Now, fill in the blanks to compute an 85% confidence interval for the average concert attendance based on 10000 bootstrapped estimates.

```
boot_attendance = estimate_attendance(__(a)__)
ci_low = np.percentile(boot_attendance, __(b)__)
ci_high = np.percentile(boot_attendance, __(c)__)
concert_interval = [ci_low, ci_high]
```

(a): (b): (c):

c) Suppose `concert_interval` comes out to $[18500, 19500]$. Which of the following statements are valid interpretations of this interval? Select all that apply.

- ☐ There is an 85% chance that the average attendance of all concerts is between 18500 and 19500.
- ☐ 85% of all concerts have an attendance between 18500 and 19500.
- ☐ If we were to make many different confidence intervals based on different samples of concerts, approximately 85% of the resulting intervals would contain the true average concert attendance.
- ☐ None of the above.

d) You are told that the data in the `"attendance"` column of `concerts` has a mean of 19000 and a standard deviation of 3000. Find the endpoints of the smallest interval which is guaranteed to contain at least $\frac{15}{16}$ of the data. Both endpoints should be given as **integers**.

$$\left[\quad \square \quad , \quad \square \quad \right]$$

e) You are now told that the data in the `"attendance"` column is normally distributed. Approximately what percentage of the data is included in the interval you gave above? Give your answer **to the nearest integer**.

Question 2

After this year's Sun God Festival, the UCSD administration wants to estimate how much students would be willing to pay for a ticket to future Sun God Festivals. They (somehow) take 500 simple random samples of 100 students each, asking them this question. They then plot a histogram showing the distribution of the mean response from each sample.

a) Which of the following statements are true? Select all that apply.

- ☐ The histogram will be approximately centered around the mean amount that all UCSD students would be willing to pay.
- ☐ The variability in the histogram is due to the fact that we resample with replacement.
- ☐ This distribution is an example of an empirical distribution.
- ☐ This distribution includes many randomly generated parameters.
- ☐ None of the above.

b) Describe in **one word** how the histogram would be different if it were instead based on 500 simple random samples of 1000 students each.

Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.